

Is Hydrogen a Solution to the Climate Change Crisis?

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Introduction:

In 1937, the German-made Hindenburg airship caught fire in a famous disaster in the United States, killing 36 people (The Economist, 2021). Ever since, there has been a negative perception of the safety of hydrogen amongst the public, despite evidence to prove that the explosion was not due to the hydrogen fuel. Hydrogen is a volatile and flammable gas but there is now huge knowledge and experience on how to keep it safe (Ibid.). With renewable energy prices coming down, hydrogen has the potential to become a major source of alternative energy. And just in time, for our planet faces an acute crisis.

At the United Nations Climate Change Conference in 2021, known as 'COP26', 197 countries agreed to a pact to limit global warming to 1.5° degrees Celsius above pre-industrial levels (UN, 2021) so as not to interfere with the Earth's climate system. This will require significant sustainable reductions in global greenhouse gas emissions created by fossil fuels with the goal of being 'Net Zero' by 2050 (Ibid.). Today, fossil fuels account for approximately 80% of the world's energy, so the emphasis is now on alternatives which includes hydrogen as a potential under-developed fuel to accelerate the energy transition.

This essay will discuss why hydrogen must be seriously considered now, exploring the advantages and disadvantages of the different types of hydrogen currently produced and how they are stored and distributed. It discusses how different countries around the world are developing their strategies for hydrogen before finally summarising where it is optimal to use hydrogen.

Background to the Climate Crisis:

In 2022 worldwide energy consumption was 604 exajoules (628×10^{18} joules) (Statista, 2023) where those energy sources are 80% "molecular," meaning coal (25%), oil (30%) and gas (22%) (Pratty, 2022). It is predicted that by 2050 this demand will rise 20% (Ibid.). This is due to people living longer, thus demanding more powered goods like cars and computers such that on a per capita basis energy use will rise. The global population itself is also predicted to increase to 10 billion (Morse, 2023), where much of this growth will be in cities which are more energy intensive. So, despite the abundance of fossil fuels that make them so cheap, there is the extra challenge of making sure that not only the current but also the rising demand for energy is net zero by 2050.

Fossil fuels are destroying the planet as they release harmful gases such as carbon dioxide, and carbon monoxide, which contribute to global warming (Donaghy, 2021), and hence increase the risk of natural disasters as witnessed this year with severe weather conditions resulting in wildfires in Europe and Canada, devastating floods in Libya, droughts in Africa and earthquakes in Turkey and Syria killing thousands of people and destroying habitats (Ibid). The sea levels are also rising from the icecaps melting where the World Health Organization have stated that this will lead to more flooding creating more water and sanitation related illnesses especially in less developed areas. This ultimately could cause mass migration as large areas become submerged and people displaced destroying economies, placing huge pressure on water and food resources and risk of potential conflicts (Ibid.).

Studies have shown that if the fossil fuel companies continue at their current rate, then climate change can cost the US government \$500 billion every year by 2090 (Donaghy, 2021). At COP26 countries pledged to limit their greenhouse emissions to 45% of current emissions by 2030 and zero emissions by 2050 which will require a huge transformation of how energy is consumed and produced (UN, 2021).

During this 'energy transition' 50% is projected to come through electrification (the use of electricity for power that replaces the use of fossil fuels), and renewable energy sources (EIA, 2022). However, this will not work for sectors in the industries that have higher energy demands, thus the case for hydrogen fuel to be used instead. Renewable energy sources, although being environmentally friendly as from abundant natural sources, are unreliable. When there is excess renewable energy that is used in the electricity grid this can then be used to make hydrogen fuel which can be stored and then used when needed.

To summarise, to combat global warming, sustainable energy must be pursued to meet the future rising energy demands, which is where hydrogen can play a huge role in the foreseeable future.

Different Types of Hydrogen Fuel:

When considering the role hydrogen might play in addressing the global Climate Crisis, one needs to understand the different types of hydrogen fuels as there are advantages and disadvantages to each. Hydrogen fuel is an alternative fuel source that is considered clean as when used its only emission is water, thus no harm to the environment (Pratty, 2022). It has a high energy density, three times more powerful than fossil fuels (Leake, 2023). However, hydrogen gas does not really appear naturally as an element on its own, it must be separated from its other elements and this process requires a lot of energy some of which is renewable, but most often today not (Pratty, 2022). One of the biggest challenges for hydrogen is its current high cost versus fossil fuel, not just in the cost of producing it, but also the infrastructure cost for storing and transporting it. There are many different types of hydrogen fuel, depending on how the hydrogen fuel is made:

Grey hydrogen, which currently equates for about 90% (Pratty, 2022) of today's total hydrogen fuel production, uses the burning of mainly natural gas which is predominantly made up of methane by separating hydrogen from the methane which has the chemical compound of CH_4 . Using a process called steam methane reforming, high temperature steam reacts with methane in the presence of a catalyst under high pressure to produce hydrogen (EIA, 2022). However, it also produces carbon monoxide and carbon dioxide. (Black and Brown hydrogen use coal instead of gas). Although today fossil fuels are one of the cheapest ways to produce hydrogen, they are the most environmentally damaging, creating 2% of global greenhouse gas emissions and so Grey hydrogen is not a long-term solution (Ibid.). The reason why Grey hydrogen dominates today is because it is cheap, the technology used to breakdown the methane molecule, Steam Methane Reformation (SMR), is well developed and competitive and the greenhouse gases are released into the atmosphere at no cost. Bloomberg NEF estimates that the average production cost in 2023 for Grey hydrogen produced from natural gas without abatement is just \$2.13/kg (Martin, 2023).

Green hydrogen is created by electrolysis which uses an electric current that is produced from a renewable energy source, such as solar power or wind turbines. Water, (H_2O) has salts and minerals in it that conduct electricity and so an

electrolyser then separates the hydrogen atom from the oxygen atom in the water (IEA, 2022). Typically, the hydrogen then goes into a compressor which increases the pressure of the hydrogen so that it can then be stored or is condensed by cooling, but crucially no carbon dioxide is produced in this process (Ibid.). Bloomberg's average estimate of cost of production for Green hydrogen is \$6.40/kg, three times that of Grey and more than double of Blue due to the cost of the green energy used and cost of equipment (Martin, 2023).

However, in the last decade the cost of solar and wind power has fallen by 87% and with further development and scaling up to meet the rising demand is expected to fall by another 80% by 2030 (The Economist, 2021). This is a long-term trend, for in the 1980s, hydrogen was considered as a 'fuel of the future' (Ibid.), but there were not enough renewable energy sources, so the only way to make hydrogen fuel would be to use fossil fuels or nuclear energy, which was more expensive and less efficient than using other energy stores at the time. Indeed, until climate change became an issue at the turn of the century, hydrogen fuel simply was not economically worth it. Whilst electrolyzers are expensive more research and development and economies of scale should also mean lower costs of production as they become more efficient. There should be similar cost benefits for the storage and transportation too, all of which should make Green hydrogen more price competitive. This is critical as the next challenge is to get those using fossil fuels to now switch to renewable energy alternatives without hurting the economics of their industry. In the drive to produce more hydrogen sustainably this should encourage companies to invest in their infrastructure such that the price of Green hydrogen could fall to \$1/kg (excluding transport and storage) (Martin, 2023) even by 2030 to those in favourable locations with good access to renewable energy (BloombergNEF, 2023). Tighter carbon pricing systems, like a carbon tax, implemented by many governments should drive up the cost of using fossil fuel which should again favour Green hydrogen (Ibid.). Green energy is still very much needed to de-carbonise the electricity grid as well as being needed to produce hydrogen sustainably.

In the meantime, Blue hydrogen will be an important part of this transition as scaling up to Green, according to senior vice-president for gas and power at Equinor, Helge Haugane (2022), "It's impossible to do in a short timeframe and that's where we believe the role of Blue hydrogen comes in."

Blue hydrogen, in terms of emissions, is in between Green and Grey hydrogen, as it is produced from natural gas using same the steam reforming process as Grey, but the main difference is that the carbon dioxide by-product is then captured and stored, typically underground or used for industrial purposes like the food and beverage industry (carbonating drinks, keeping food cold during transportation) (Fowler, 2021). There are many mixed opinions on Blue hydrogen from being the most promising cost-effective solution in the near term to critics believing that it is just another "climate horror story" (Thomas, 2022), allowing fossil fuel producers and gas infrastructure owners to continue to use their products far into the future. For Blue hydrogen compared to Grey, the capture and storage management of the carbon dioxide by-product, adds a further cost. Bloomberg NEF estimates this average cost including carbon capture at \$3.10/kg (Martin, 2023).

The argument for Blue hydrogen is aided by companies like MCI Carbon based in Australia who have developed a breakthrough technology called mineral carbonation which involves building carbon plants that capture CO₂ emissions and convert them profitably into solid carbonate materials that can then be used in construction, consumer and

manufactured products. They showcased their technology at COP26 winning the 'No.1 Clean Energy Startup' award and are now looking to scale up on a more industrial level (MCi, 2021).

The following hydrogen types also exist and could have the potential to be materially significant, but they are much more recent ideas and discoveries and hence are currently very under-developed.

Emerald hydrogen is a relatively new development that heats waste materials in a container, without oxygen, to produce energy without the combustion, and hence zero emissions. HiiROC, based in Yorkshire, is an example of a company developing this where they convert hydrocarbons under high pressure using plasma torches to split methane into hydrogen and carbon (HiiROC, 2019). This thermal plasma technology not only uses much less energy, but it can easily be made near the point of use thereby cutting transport and storage costs and its by-product is solid carbon which is then used for car tyres, rubber plastics and the steel and concrete industry (Ibid.). Innovations like these could be game changing.

White hydrogen is naturally occurring hydrogen that is found in deep underground deposits formed by fracking, but no major large deposits had been found. Until however, in 2023, Jaques Pironon and Philippe De Donato whilst originally looking for fossil fuels underground using a probe to research methane concentrations in the soil in France, discovered that about 200 meters down, there were low concentrations of hydrogen, and as they went deeper, the concentration increased, such that by 1250 meters down it was at 20% (Paddison, 2023). This indicated that there was a hydrogen reservoir below, and estimations showed that the reservoir could contain between 6 million and 250 million metric tonnes of pure hydrogen (Ibid.). Ellis, a geochemist with the US Geological Survey, having thought not viable 4 years ago, now believes there could be large reserves of hydrogen, but the challenge is finding and extracting it economically as many of these reservoirs are likely to be offshore or just too deep (Ibid.).

In conclusion, there are many methods to produce hydrogen fuel. However, in order to cut emissions, more focus is needed on producing Green hydrogen, using sustainable energy sources, as opposed to Grey hydrogen, that is damaging to the environment. It will take time to build up sufficient renewable electricity resources, so in the meantime Blue hydrogen is a potential solution, if it can be successfully captured and stored, particularly if developments like carbonation can then use the carbon dioxide by-product. White and Emerald hydrogen could also be very meaningful but requires more research.

The Challenges of Hydrogen Storage and Distribution:

The wind does not blow, and the sun does not shine at the precise time that energy is needed, so it must be stored. A lot can be plugged directly into the grid, but often the grid does not need it and so we have to find a way to store this surplus for future use. Often this energy can be used to recharge batteries, but batteries run flat and are not appropriate for longer term storage (for example in the winter), which is where hydrogen matters, as it can store energy for long periods of time.

However, hydrogen is complicated to store due to its low volumetric energy density meaning it needs large storage containers for the same amount of energy versus alternatives, though many solutions have now been developed to

combat this depending on its purpose. It can be compressed into a liquid in high pressure tanks or cryogenic containers (Leake, 2023). For example, hydrogen refuelling stations, which whilst saving space unfortunately use energy intensive methods thereby reducing the energy efficiency of hydrogen given that it requires energy to produce it as well now as compressing and storing it.

Hydrogen can also be stored like natural gas underground. Mitsubishi Power and Magnum Development are working on a project in Utah which promises to build a hydrogen storage facility for 1000 megawatts of clean power by keeping the hydrogen in salt mines scheduled to be in operation by 2025, (CNBC, 2020) making it the largest clean energy storage system in the world. A large electrolysis system will be built above and will produce the hydrogen fuel to then be stored in the salt system underground which then can be used when the grid needs it, and not when it is first produced. It will be built next to the last coal powered power plant that provides power to California, scheduled to retire in 2025, and it will be converted to use turbines powered by a mixture of natural gas and hydrogen fuel, with the ultimate goal to use 100% hydrogen fuel by 2045. Phase 1 will have 150000 megawatts of energy storage, which is enough to fuel 30% of 840 megawatt powerplant which is then enough to supply 1/7 of the needs of Los Angeles (Ibid.).

Hydrogen can also leak very easily and is highly flammable, and can corrode metals, meaning that great safety precautions must be made for both storage and transportation. This whole process is expensive but not impossible. For similar reasons, distribution is also a challenge (Forbes, 2021). There are three main methods of the distribution. The first is pipelines which are the cheapest way to deliver large volumes of hydrogen fuel and can be distributed over thousands of miles, but it has a limited capacity, although some of the existing gas pipe infrastructure in place can be adapted for hydrogen. The second method to distribute hydrogen fuel is via high-pressure tube trailers, where hydrogen fuel is compressed and then transported by truck, railcar and ship, but this is expensive, and so the hydrogen fuel cannot travel far (typically no more than 200 miles). The final way is through liquefied hydrogen tankers using cryogenic liquification, where the hydrogen is cooled to become a liquid, allowing it to be transported both further and more efficiently (Alternative Fuels Data Centre, 2023). Again, this is also an expensive process, and it has to be dealt with at a sufficiently high rate when being used otherwise it can evaporate (US. Department of Energy, 2023).

To conclude, due to the flammability of hydrogen, many safety precautions have to be made when both storing and distributing hydrogen fuel, which also raises the cost. However, there are already many methods that are used to increase the safety of handling hydrogen fuel, and hence allowing it to remain a viable solution to the climate change crisis.

The Importance of Regional Government Support:

The magnitude of investment needed to change the energy infrastructure for hydrogen is vast and so governments will have to be very active in encouraging and funding this. Although hydrogen is a highly dense powerful clean fuel, it could be deemed inefficient because energy is needed to produce it, so should only really be used where direct electrification is not feasible. Thus, it is more attractive to industrial processes that need a chemical reaction such as production of steel, cement or chemical manufacturing or long-range transport. Hydrogen is also used today for feedstock and producing fertiliser, but only about 70 million tonnes, a fraction of what would be needed for the above (Deign, 2019). The USA, Europe and Asia recognise that hydrogen is needed in significantly large quantities for these

industries to decarbonise and so in the interim that demand will have to be fulfilled with Blue hydrogen due to it being cheaper. But this is important as it will help accelerate the investment in infrastructure for these industries to change and thereby to cut their emissions, before finally hopefully switching Green hydrogen. For example, BP Teeside is a project that aims to be one of the largest Blue hydrogen production facilities in the UK, producing up to one gigawatt of Blue hydrogen (20% of the UK's hydrogen targets) by 2030 and supporting the development of the region as the UK's first transport hub (H2 Teeside, 2023).

The International Energy Agency estimates approximately \$150 trillion of capital is needed for the energy transition to net zero, with the majority to electrify systems and the electrification of activities. They predict that by 2050 electricity will account for 50% of the global energy needed, up from 20% today (IEA, 2021), but that still leaves a lot to invest for hydrogen. What happens in each country will depend on the role played by their government which varies greatly and so it is necessary to look at hydrogen strategies on a regional basis.

Europe

The UK and Europe already use carbon pricing which is essentially a tax to the manufacturers on carbon emissions to produce goods. The aim is that this tax is likely to rise such that this plus the actual cost of the fossil fuel used will be more expensive making hydrogen fuel more competitive on price. However, the Hydrogen Council reported in 2023 that whilst over 1,000 clean hydrogen projects globally have been announced that would need a total investment of \$320bn by 2030 (Collins, 2023) of which \$150bn is for scaling up to gigawatt capacity for Green and low-carbon production. Yet only \$29bn (Ibid.) of investment has been committed so Governments need to do more to promote Hydrogen.

Europe has opted for a regulatory framework for its hydrogen market using a 'hydrogen accelerator' concept to produce 10 million tonnes and import 10m tonnes of renewable hydrogen into the EU by 2030, twice that of today (Collins, 2023). This 20 million is an ambitious target and will add extra costs on industry as they are forced to switch especially as the electricity used must be Green, where the grid is often a mix. The EU has also launched a new European Hydrogen bank to help promote and support these investments into hydrogen. France has a €100 million investment plan for the necessary technology to make France the world leader in hydrogen (Deign, 2019); whilst Germany has €100 million to invest in 20 research laboratories for hydrogen technology and development where they are already advanced versus their peers (Ibid.). In the UK, the government made a £12 billion plan to use 4 gigawatts of renewable offshore wind energy to produce Green hydrogen (Ibid.) and HyNet is responsible for the creation of the UK's first carbon capture and storage infrastructure.

Most countries in Europe are working towards having a green hydrogen economy, with investments made to build renewable sources and developing new technology, and a tax to decrease their carbon footprint.

Asia and Australasia

In 2017, Japan, a country that is reliant on importing fossil fuels and with many of its nuclear reactors out of action since the 2011 Fukushima disaster, was one of the first countries to format a plan for hydrogen. This included the aim for 800,000 hydrogen powered cars on the roads by 2030 (ETEnergyworld, 2023) pioneered by their leading car

manufacturers Toyota and Honda. However, even with subsidies these cars and the installation and upkeep of the refuelling stations are expensive. Another aim was to have a residential fuel cell programme called 'Enefarm' to provide energy for over five million homes by 2023 (Ibid.), but they are well behind their target. Their strategy was based on mainly Grey and Blue hydrogen with less focus on renewable sources unlike other countries. Japan is now revising its strategies on increasing hydrogen fuel supplies to six times by 2040, as well as considering using ammonia to reduce carbon emissions and importing more Green hydrogen energy.

In Australia, the Government is investing heavily in developing Green energy given its abundance of natural resources including its ideal climate for solar and wind power, approximately 3% of their land mass is appropriate (Australian Government, 2022). Indeed \$1.3billion Australian dollars has been committed to accelerate the development with the largest number of projects of any country as it is expected to be a huge net exporter of hydrogen given its proximity to Asia (Ibid). For example, Australia and Japan recently made a deal for Australia to produce brown liquid hydrogen and transport it to Japan for low-emission use. Australia will have eight industrial hubs which will collate hydrogen from the various producers using existing infrastructure like pipelines and storage tanks. They also intend to invest in hydrogen refuelling stations, particularly on freight routes and support the decarbonization of existing industries and public infrastructure. Their strategy is adaptive, but their ultimate goal is to be a major player in this market.

China today is the largest producer of hydrogen, some 33 million tons, but the majority is produced using fossil fuels. Their long-term plan announced in 2022 implies demand of 35Mt by 2030 but only 100-200 tons will be from renewable sources (Nakano, 2022). As part of its plan to be net zero by 2060, China is mainly focusing on decarbonizing transportation by using both Green hydrogen and electric vehicles with the target of one million fuel-cell vehicles by 2030 (Ibid.).

India is not as focused as the rest of the world to become net zero, with plans to be energy independent by 2047, and net zero by 2070 (india.gov.in, 2023).

In Asia, Australia has developed an advanced plan to produce and export green hydrogen to countries like Japan who are more focused on the research and development for the use of hydrogen. The less developed countries, with later Net Zero timescales, are still evolving their plans.

North and South America

In August 2022, the US signed an Inflation Reduction Act (IRA) which offers producers of Green hydrogen a tax credit of up to \$3/kg. This is like a subsidy that should make Green hydrogen much cheaper than Grey to produce which should accelerate conversion to Green. However, despite being potentially worth up to \$100bn it is only for the next ten years, which is a short time frame for these industries to make such large upfront investments thereby some risk.

Eleven countries in Latin America plan to produce large volumes of low-carbon hydrogen which they can then export to global markets (IEA, 2021) as well as taking steps to use electrification and renewable energy. In Chile and Peru, the use of hydrogen in mining can have a huge positive impact on the countries' carbon emissions (Ibid.).

Overall, there are several different hydrogen strategies taken by countries around the world which are influenced by many factors including their economic status, current carbon footprint, access to potential renewable sources and research and development facilities. It could be argued that the developed countries will need to lead this, but for every country there must be Government support to help regulate, fund and encourage investment into hydrogen use and infrastructure where most suited.

What energy uses is hydrogen fuel most suited to?

As with types of hydrogen fuel, when considering how hydrogen might help solve the Climate Crisis, one must consider the different applications of hydrogen fuel as there are advantages and disadvantages to each. As discussed, hydrogen is not as efficient to produce as electricity and so not always appropriate, therefore focus should be on sectors that are not only the worst polluters but also need the advantages of hydrogen over electricity. While all sectors need to decarbonise, transport, heating and industry account for approximately 77% (Kavanagh, 2022) of global emissions and so one needs to look at each of them in turn.

Heavy Industry:

The biggest demand for hydrogen fuel will undoubtedly be from the heavy industry, particularly the steel and cement industries that each account for 8% of fossil fuel emissions as electricity does not work due to the extreme heat needed during production (Pfeifer, 2021). In the production of 'new' steel hydrogen fuel is the chemical reductant that rips oxygen out of iron ore in a blast furnace, producing high quality steel. Whilst in the steel industry there may be a shift more to the use of Electric Arc Furnaces over Blast Furnaces as seen in America where 70% of steel is made this way, this method is reliant on scrap metal (for example used cars) and is just not appropriate for many developing countries (Ibid.). Thyssenkrupp's Duisburg steel plant in Germany produces around 11 million tonnes of steel a year and 20 million tonnes of carbon dioxide a year (5% of Germany's CO₂ emissions). Given the EU regulation mentioned above, Thyssenkrupp needs to decarbonise quickly and they're trialling using hydrogen as a reducing agent in blast furnaces which if works efficiently, can bring their carbon emissions down by 20% (Ibid.). The Swedish steel maker SSAB is working on making fossil fuel free steel using HYBRIT technology, using recycled steel as the raw material in an electric furnace powered by sustainably sourced electricity (SSAB, 2023).

Using Green, or even Blue hydrogen to fuel the heavy industry, particularly steel and cement industries, will have a huge positive impact on our climate, and it is essential that we act on this.

Transport:

The global head of fuel decarbonisation at Trafigura, Bach Nielson (2023) claimed that, "the cleanest, most realistic transport fuels of the future are hydrogen-based fuels, including Green ammonia," and that "technology is there and ready. We now just need regulation" (Dempsey, 2021). In practice, the applicability of hydrogen will vary from high in some sectors like shipping to low in sectors like automotive.

Electric batteries already dominate the car industry as they are cheap to charge, and it is much easier to install the infrastructure for electric charging stations despite hydrogen being lighter and quicker to charge (Miller, 2021). The

Volkswagen chief executive, Herbert Diess (2021), commented, “you won’t see any hydrogen usage in cars” because “the physics behind it is so unreasonable”. Indeed, apart from Honda, Hyundai & Toyota where their cars are twice the price, most major car companies have given up on hydrogen (Ibid.). There are, however, possibilities where hydrogen could be a compelling option for transportation.

US auto producer, Nikola, is developing a hydrogen fuelled truck as, the larger the vehicle, the more advantageous it is to use over large heavy lithium batteries, though more refuelling stations would be needed. Similar technology can be used for buses and trains and with shipping producing about 3% of global greenhouse gas emissions the UN agency want to at least halve its impact by 2050 (Dempsey, 2021). Kristoffersen, chief executive of Norway’s Torvald Klaveness (2021) said, “there is no question whether hydrogen will be the energy carrier of shipping in 2050. The question is how do you produce it and which form do you use it as a carrier?” Since hydrogen would have to be stored in cryogenic tanks (in liquid form below -253°C) which takes up a lot of space, it is not ideal for cargo ships, so the shipping industry is leaning towards the idea of using ammonia (NH_3) as fuel, even though it has an unpleasant smell and is toxic, it has nearly twice the energy density of liquid hydrogen (Ibid.).

Although hydrogen fuel is already established for rocket fuel, the aviation industry which is one of the most polluting forms of transportation contributing 3.5% to global warming (IBERDROLA, 2023), is also seeing potential to use hydrogen. In September 2023, a hydrogen powered plane, developed by the German company H2FLY, successfully flew from London to Rome (Leake, 2023). The plane used liquid hydrogen to generate power for electric motors that turned propellers, as liquid hydrogen is lighter than batteries and takes up less space. Josef Kallo (2023), the co-founder of H2FLY said that they are now “looking to scale up their technology for regional aircraft and other applications” and that “In just a few years, hydrogen-electric aircraft are expected to be able to transport 40 passengers at a time over distances of up to 2000 kilometres.”

Hydrogen fuel is currently not essential for short-distance transport vehicles, such as cars as there are other renewable solutions like electric cars or hybrids. For longer distances for trucks, ships and planes, hydrogen fuel is a suitable solution to cut down carbon emissions.

Domestic Heating:

This sector has poor prospects for hydrogen usage. In the UK, for example, where 20% of emissions come from the way we heat homes and businesses (Kavanagh, 2023) David Cebon, professor of mechanical engineering at Cambridge university stated that, “In the UK, heating homes with Green hydrogen would use approximately six times more renewable electricity than heat pumps” (Fisher, 2022), so it is not worthwhile investigating further despite domestic hydrogen boilers already having been developed. Indeed, Germany is choosing not to switch to hydrogen fuel for heating homes yet as they are implementing heat pumps as a short-term alternative to natural gas.

Jason Rosenow, a European programme director at RAP stated that, “the good news for the Green hydrogen industry is we need lots of it, but the priority should be where it’s clear that it’s a good use of hydrogen”. He wrote a review of 30 worldwide studies where he concluded that hydrogen used in domestic use is less economic, less efficient and more resource intensive and that there are other alternatives such as heat pumps, solar thermal and district heating systems

that would be more appropriate for domestic use, as hydrogen should be focused on areas with a larger environmental impact (Kavanagh, 2023).

To summarise, Hydrogen can be used in many sectors, however it must be prioritised in the areas with the highest carbon footprint where electricity is not appropriate. As of now, the focus of using hydrogen fuel is within the heavy industry and long-distance travel such as shipping or the aviation industries as opposed to domestic heating and cars.

Conclusion:

COP26 set out ambitious climate change goals and the use of fossil fuels is expected to peak in 2030, but consumption of oil and natural gas is forecast to plateau after that when it needs to decrease quickly. Hydrogen will not be *the* answer to the Climate Change crisis, but it can definitely be part of the solution. It will take considerable investment by the public and private sectors. The focus must be on producing hydrogen sustainably with Green hydrogen being the ultimate goal. In the meantime, the production of Blue hydrogen is critical in this transition, whilst recognising that there are early-stage developments in areas like mineral carbonation, emerald hydrogen that could be game changing or finding meaningful reserves and methods of extraction of White hydrogen.

Governments must play a major role to accelerate the energy transition for hydrogen in conjunction with renewables since they depend on one another. There needs to be more incentives, regulations and finance. In determining the efficient use of sustainable energy, the considerations of where hydrogen should actually be used means that its use is only appropriate for sectors where electrification is unfeasible, such as the heavy industry, long distant transport and long-term storage of energy, but not for heating homes or driving cars. Producing, storing and distributing hydrogen sustainably and cost effectively as well as getting industries to switch to now using hydrogen instead of fossil fuels will require significantly more global research and development.

Most importantly, action needs to be taken quickly. The effect of the Hindenburg disaster and the high cost of renewables meant that it has not been feasible to deploy the element at scale until now. But now that it can be, it must. Surface temperatures are already at 1.1°C above pre-industrial levels today, and so there is no time to delay.

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